Provability as membership in every maximal extension

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the statement's final clause '(including when Gamma is empty, in which case...)' is inside the span as printed.

Editorial note. The label cor:provability-characterization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Lindenbaum's lemma. Same part, \ollabel{thm:lindenbaum} (source line 983, own card). Line 1108: the converse case extends the consistent set to a complete one omitting A.
- Monotonicity and reflexivity of Sigma-derivability. Proof theory part: structural properties of the derivability relation. Lines 1101-1102: the forward direction's two quick derivations from Delta.

Corollary 0.1. $\Gamma \vdash [\Sigma]A$ *if and only if* $A \in \Delta$ *for each complete* Σ*-consistent set* Δ *extending* Γ *(including when* $\Gamma = \emptyset$*, in which case we get another characterization of the modal system* Σ*.)*

Proof. Suppose $\Gamma \vdash [\Sigma]A$, and let Δ be any complete Σ-consistent set extending Γ. If $A \notin \Delta$ then by maximality $\neg A \in \Delta$ and so $\Delta \vdash [\Sigma]A$ (by monotonicity) and $\Delta \vdash [\Sigma]\neg A$ (by reflexivity), and so Δ is inconsistent. Conversely if $\Gamma \vdash /[\Sigma]A$, then $\Gamma \cup \{\neg A\}$ is Σ-consistent, and by Lindenbaum's Lemma there is a complete consistent set Δ extending $\Gamma \cup \{\neg A\}$. By consistency, $A \notin \Delta$. □